

SAMUEL K. RYAN

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Certified Data Center Design Professional (CDCDP)

SUMMARY

Operations Engineer and Certified Data Center Design Professional with 14 years operating mission-critical power and cooling across a global IT portfolio - and, uncommonly, the developer who builds the monitoring, alerting, and automation software behind it. Pairs hands-on critical-facilities expertise (UPS, generators, switchgear, precision cooling) with production .NET/SQL/Python development and sole ownership of the Nlyte DCIM platform. Known for replacing manual and vendor processes with owned, automated tooling.

CORE SKILLS

Critical Facilities: UPS, generators, ATS and switchgear, precision cooling, Tier reliability standards, data center design (CDCDP)

Monitoring & BMS: DCIM (Nlyte Asset Optimizer / NAO, Nlyte Energy Optimizer / NEO, Fieldview); Niagara-based BMS and environmental monitoring (EnviroNET); centralized alerting and dashboards

Protocols: SNMP, Modbus, BACnet; device polling, integration, and data normalization

Software & Data: C#/ .NET (Windows services, ASP.NET web apps), MSSQL / T-SQL (stored procedures, Dapper), Python, PowerShell, REST and GraphQL APIs, Git; AI-assisted development

Delivery: Capital project management, vendor and contractor management, budgets and POs, incident and on-call response, root-cause analysis, advanced Excel reporting

PROFESSIONAL EXPERIENCE

Asurion - Operations Engineer IV, Critical Systems Support

July 2012 - Present | Nashville, TN

- Design and maintain the critical-facilities platform (generators, UPS, precision cooling, and associated switchgear), targeting 99.9999% power and cooling availability across a global portfolio of data centers, call centers, and COLOs.
- Architected and shipped, as sole developer, a self-hosted monitoring and incident-management platform in .NET that replaced a three-vendor alarm pipeline (SNMP / BigPanda / GoAlert): per-site incident grouping, noise suppression (flapping detection and maintenance windows), on-call paging, a web acknowledgment app, and automated email reporting - bringing paging logic fully in-house.
- Integrated 275+ power and environmental monitoring cards (APC and Vertiv) over SNMP, Modbus, and BACnet - building device templates and polling configuration; fleet-wide capture of utility-to-generator transfers and on-battery events, with noise filtering.
- Sole administrator of the Nlyte DCIM platform (asset and energy/monitoring) across 70+ global sites; own asset lifecycle, polling and monitoring configuration, and cross-system reconciliation.
- Develop and maintain the MSSQL behind the DCIM (stored procedures, complex queries, data validation); diagnosed and resolved a data-warehouse ETL failure that had silently broken reporting.
- Delivered capital projects end to end - UPS, generator, and precision-cooling (CRAC) replacements; new call-center and IT-space build-outs; and large-scale decommissions - across multiple countries, managing vendors, budgets, and schedules.
- Led a Tier II to Tier III electrical-infrastructure resiliency upgrade for a critical site.
- Build Python tooling and automated telemetry/Excel reporting for power consumption and capacity analysis (used for Energy Star submissions).
- Design installation layouts and upgrade plans from engineering drawings and company data center standards; first responder for electrical and mechanical emergencies.

Travis Electric - Journeyman Electrician

May 2007 - July 2012

- Installed and maintained electrical wiring, equipment, and fixtures to national and local code across commercial and industrial projects.
 - Planned work from schematics and blueprints; operated hand and power tools, hoists, precision instruments, and electrical test equipment.
 - Completed service reports and warranty documentation; maintained safe, orderly worksites and strong customer relationships.
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EDUCATION

B.S., Mathematics - University of Tennessee at Martin

CERTIFICATIONS

- Certified Data Center Design Professional (CDCDP)
 - OSHA 10 (Construction)
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COMMUNITY

- Youth baseball coach - Smyrna Baseball League and MBSA (Murfreesboro), TN. Coach rec and travel teams (USSSA and Play Local tournaments); lead practices, player development, and game-day management.
- Volunteer, local church.